Oral Exam Syllabus

Avinash Iyer

October 2026

The oral exam for Avinash Iyer will be administered by Professors Ben Hayes and Mikhail Ershov on December 4, 2026 focused on Functional Analysis and Geometric Group Theory. The exam format will be as follows.

Presentation: "The Nonstandard Analysis Proof of Gromov's Theorem:" a 30 minute chalk talk based on the paper "Gromov's Theorem on Groups of Polynomial Growth and Elementary Logic" by van den Dries and Wilkie, covering ideas from nonstandard analysis and a broad outline of how they're used in the proof of Gromov's Theorem.

After the presentation, the question-and-answer portion of the exam will take place. Each of the following topics will be examined with the indicated time delineation.

Functional Analysis Topics: (approximately 60 minutes)

- Locally Convex Spaces, topics covering Chapter IV §1–§3 of Conway's *A Course in Functional Analysis*:
  - definitions and examples;
  - metrizability and normability;
  - Hahn–Banach separation (and applications).
- Weak topologies and extremal structure, topics covering Chapter V §1–§5 and §7–§8 of Conway's *A Course in Functional Analysis*:
  - dual spaces of locally convex topological vector spaces;
  - weak compactness, Banach–Alaoglu Theorem;
  - reflexivity and separability;
  - Krein–Milman theorem.
- C^*-algebras and von Neumann algebras, topics covering Chapter VIII §1–§3, §5 of Conway's *A Course in Functional Analysis* and Sections 4.6 and 4.7 of Pedersen's *Analysis Now*:
  - Gelfand Transform, Gelfand–Naimark theorem, and continuous functional calculus;
  - ordering properties and their relation to the spectrum;
  - GNS construction;
  - von Neumann's double commutant theorem;
  - σ-topologies and the predual of a von Neumann algebra.
- Spectral theory for normal operators, topics covering Chapter IX §1–§8 of Conway's *A Course in Functional Analysis* and Sections 4.4–4.6 of Pedersen's *Analysis Now*:
  - spectral measures and projections;

- Borel functional calculus (and applications);
- the characterization of abelian von Neumann algebras.

Geometric Group Theory: (approximately 45 minutes)

- Relations between groups and their action on graphs, topics covering Chapters 3 and 4 of Löh's *Geometric Group Theory*:
  - Cayley graphs and word metrics;
  - actions of groups on trees.
- Quasi-isometry, topics covering Chapter 5 and 6 of Löh's *Geometric Group Theory* and sections 11.12–11.13 of Ceccherini-Silberstein and d'Adderio's *Topics in Groups and Geometry*:
  - definition of quasi-isometry of metric spaces and groups, quasi-geodesics;
  - Švarc–Milnor lemma and its topological formulation (statement and applications);
  - dynamic criterion for quasi-isometry (set-theoretic coupling and topological coupling);
  - asymptotic cones (quasi-isometry induces bilipschitz equivalence of asymptotic cones);
  - quasi-isometry invariance of growth rate.
- Amenability, topics covering Chapter 9 of Löh's *Geometric Group Theory* (for more geometry-oriented results) and Brown and Ozawa's *C*-Algebras and Finite-Dimensional Approximations* (for more functional analysis-oriented results):
  - definition via means, elementary inheritance;
  - sufficient criteria (polynomial growth, solvability);
  - equivalent criteria:
    - existence of Følner sequences;
    - approximation properties (almost-invariant vectors for left-regular representation, approximately invariant probability measures on G);
    - fixed-point property;
    - existence of a net $(\varphi_i)_i$ of finitely supported positive definite functions on Γ converging to 1 pointwise;
    - $C^*(\Gamma) = C^*_\lambda(\Gamma)$.
- Soficity, topics covering content from Pestov's *Hyperlinear and Sofic Groups*, Pestov and Kwiatowska's *An Introduction to Hyperlinear and Sofic Groups*, and Ceccherini-Silberstein and Coornaert's *Cellular Automata and Groups*
  - equivalent definitions:
    - almost-homomorphism into symmetric groups;
    - sofic approximations and metric ultraproducts;
  - inheritance properties and prominent examples (amenable groups, residually finite groups).

References

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